PRESENTATION WALKTHROUGH

Energy AI Hackathon 2026

Oil Production Prediction Using Machine Learning

Team Brain Oil
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HOW TO USE THIS GUIDE

This walkthrough follows the official 6-slide hackathon template. Each section corresponds to one slide. Use this as your speaking notes. You have less than 5 minutes to present.

Symbol	Meaning
SAY:	What to say out loud to the audience
SHOW:	What to point to or emphasize
KEY POINT:	The main takeaway for that slide

PRESENTATION TIMELINE (~5 minutes)

Slide	Topic	Time
1	Title & Team Introduction	15 sec
2	Executive Summary (4 questions)	1 min
3	Workflow Overview	1 min
4	Key Modeling Decisions	1 min
5	Results and Discussions	1 min
6	Feedback & Thank You	45 sec

SLIDE 1: TITLE

SAY: "Good morning! We are Team Brain Oil, and we're presenting our machine learning solution for predicting oil production."

SHOW: Point to team names on the slide.

KEY POINT: Introduce yourselves confidently and quickly.

SLIDE 2: EXECUTIVE SUMMARY

SAY: "Let me quickly answer the four key questions..."

- **The Problem:** Predict 3-year cumulative oil production for 12 new wells with 100 uncertainty realizations.
- **Our Solution:** Ridge Regression with alpha=1.0, StandardScaler normalization, stepwise feature selection, and Bagging for uncertainty.
- **What We Learned:** Simple models beat complex ones for small datasets. Porosity is the #1 predictor.
- **Our Results:** Test R^2 = 0.9905 (EXCELLENT), RMSE = 4.7% error.

KEY POINT: We achieved EXCELLENT results by matching model complexity to data size.

SLIDE 3: WORKFLOW OVERVIEW

SAY: "Our workflow has 7 key steps..."

1. Data Loading - 71 training wells, 12 test wells, sand proportion map
2. MICE+CART Imputation - Fill missing values at depth level (Van Buuren 2018)
3. Aggregation - Multiple depth rows become one row per well (mean, std, min, max)
4. Feature Engineering - 105 features including RQI, FZI, spatial features
5. Feature Selection - Correlation filter then stepwise selection (105→10)
6. Model Training - Ridge Regression with StandardScaler
7. Uncertainty - Bagging Ensemble with 100 estimators

KEY POINT: Every step is research-backed with peer-reviewed citations.

SLIDE 4: KEY MODELING DECISIONS

SAY: "Two key insights drove our success..."

Why Ridge Regression?

- With only 71 training wells, complex models overfit
- Hastie et al. (2009): Regularized linear models beat trees for small n
- Ridge R^2=0.99 vs Random Forest R^2=0.85 vs XGBoost R^2=0.82

Why Porosity (ϕ) Matters Most?

- Porosity directly measures how much oil the rock can store
- It appears in the fundamental OOIP equation
- Domain experts always prioritize porosity over permeability

KEY POINT: Simple model + domain knowledge = winning combination.

SLIDE 5: RESULTS AND DISCUSSIONS

SAY: "Our model achieved EXCELLENT performance by industry standards..."

Metric	Our Result	Industry Benchmark
Test R^2	0.9905	≥0.93 = Excellent
CV R^2	0.9539 ± 0.0456	Stable
RMSE	1.57M BBL (4.7%)	<10% = Excellent

- Predictions for all 12 wells (72-83) with 100 realizations each
- Production range: ~15M to ~47M BBL over 3 years

KEY POINT: R^2 = 0.99 means our model explains 99% of production variation.

SLIDE 6: FEEDBACK

SAY: "To wrap up, here's what we learned and our feedback..."

What We Learned:

- Simple models win for small datasets
- Domain knowledge (porosity priority) improves results
- MICE+CART is the gold standard for imputation

What We Liked:

- Real-world petroleum engineering challenge
- Excellent workshop materials from Prof. Pyrcz and Prof. Foster

SAY: "Thank you for your attention. We're happy to answer questions."

KEY POINT: End confidently and invite questions.

QUICK REFERENCE CARD

Setting	Value
Model	Ridge Regression
Alpha	1.0
Normalization	StandardScaler (CRITICAL)
Sand Map	Smooth 3x3
Feature Selection	Stepwise (105 → 10)
Uncertainty	Bagging Ensemble (100)
Test R^2	0.9905 (EXCELLENT)
RMSE	1.57M BBL (4.7%)

ANTICIPATED Q&A;

Q: Why not use Random Forest or XGBoost?

A: With only 71 training wells, complex models overfit. Ridge's R^2=0.99 beat RF's 0.85 and XGB's 0.82.

Q: Why is porosity the most important feature?

A: Porosity directly measures storage capacity. It's in the fundamental OOIP equation: OOIP = 7758 × A × h × ϕ × (1-Sw) / Bo.

Q: How did you handle missing data?

A: MICE + CART imputation at the depth level before aggregation, per Van Buuren (2018) recommendations.

Q: How confident are you in these predictions?

A: Very confident. R^2=0.99 with only 4.7% error. The 100 realizations capture uncertainty via Bagging Ensemble.